

PIONEER JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2015

**Candidate
Name**

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Civics

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Index Number

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H2 BIOLOGY

9648 / 02

17 Sept 2015

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Read the instructions on the answer sheet very carefully.

This document consists of 2 sections:

Section A:

Answer **ALL** questions.
Answers are to be written in the spaces provided.

Section B:

Answer **ONE** question.
Write your answers on the separate writing papers provided.

For Examiner's Use	
Structured Questions	
1	/ 12
2	/ 12
3	/ 12
4	/ 12
5	/ 10
6	/ 12
7	/ 10
8 or 9	/ 20
Total	/ 100

Do not open this booklet until you are told to do so.

Section A: Structured Questions [80 marks]

Answer all the questions in the space provided.

Question 1 [12 marks]

Collagen is the main structural protein of the various connective tissues in animals. As the main component of connective tissue, it is the most abundant protein in mammal, making up from 25% to 35% of the whole-body protein content. **Fig. 1.1** shows the structure of a collagen fibre and collagen fibrils.

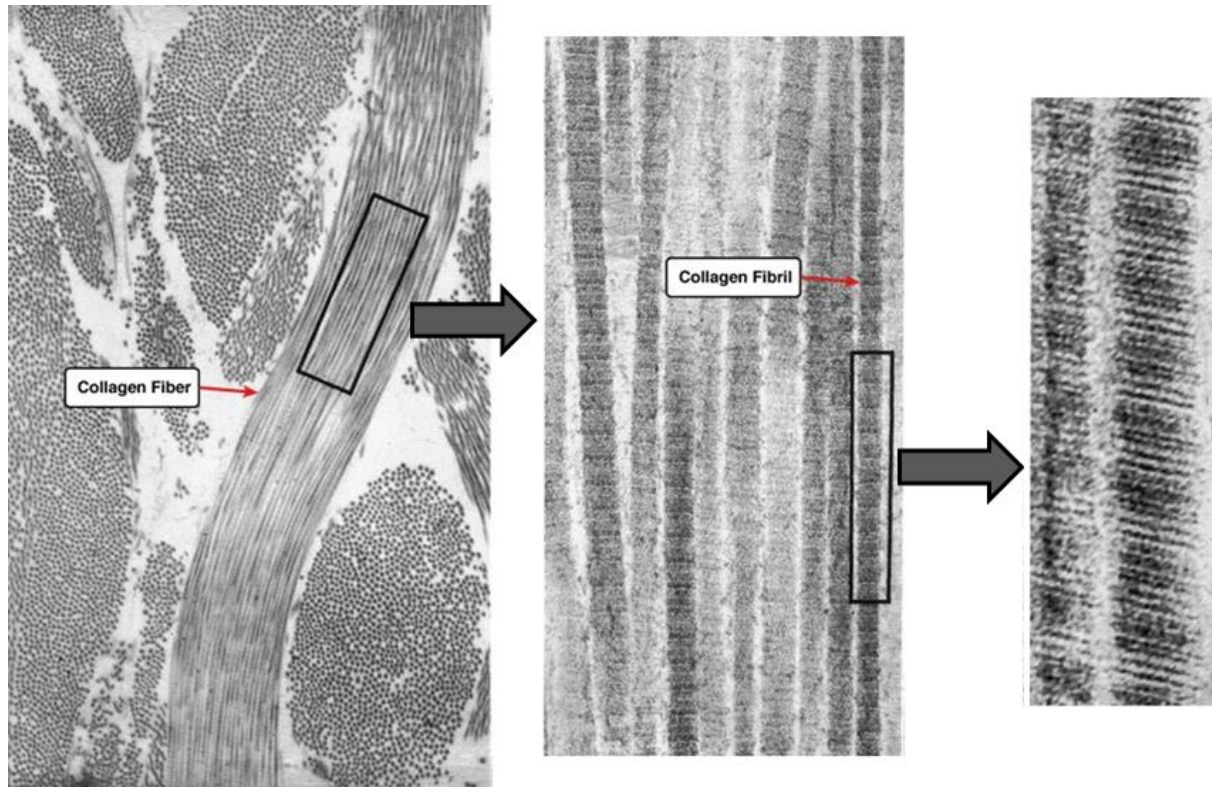


Fig. 1.1

(a) (i) Describe the primary structure of a collagen polypeptide. [2]

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(a) (ii) Explain why the collagen fibril appears as a banded structure. [1]

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Besides structural proteins in the human body, there are other proteins that play important roles in cellular processes too. An example would be pepsin, which digests proteins ingested so that amino acids can be taken up by cell for further protein synthesis. **Fig. 1.2** shows the structure of human pepsin.

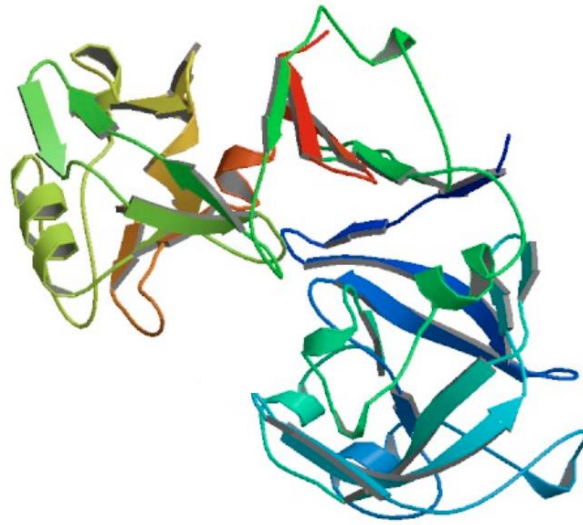


Fig. 1.2

(b) With reference to **Fig. 1.1** and **Fig. 1.2**, describe two ways how the structure of a collagen fibre differs from that of an enzyme such as pepsin. [2]

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Fig. 1.3 shows the molecular structure of a dipeptide, which is formed from two amino acids.

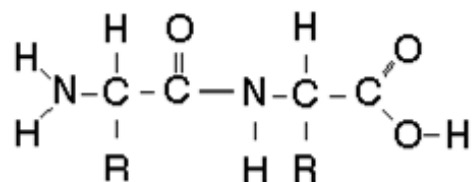


Fig. 1.3

- (c) Describe how the bond holding the two amino acids together may be broken to release the two amino acids. [2]

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Another example of proteins with a crucial role in maintaining stable intracellular conditions would be transport proteins embedded in the cell surface membrane (**Fig. 1.4**).

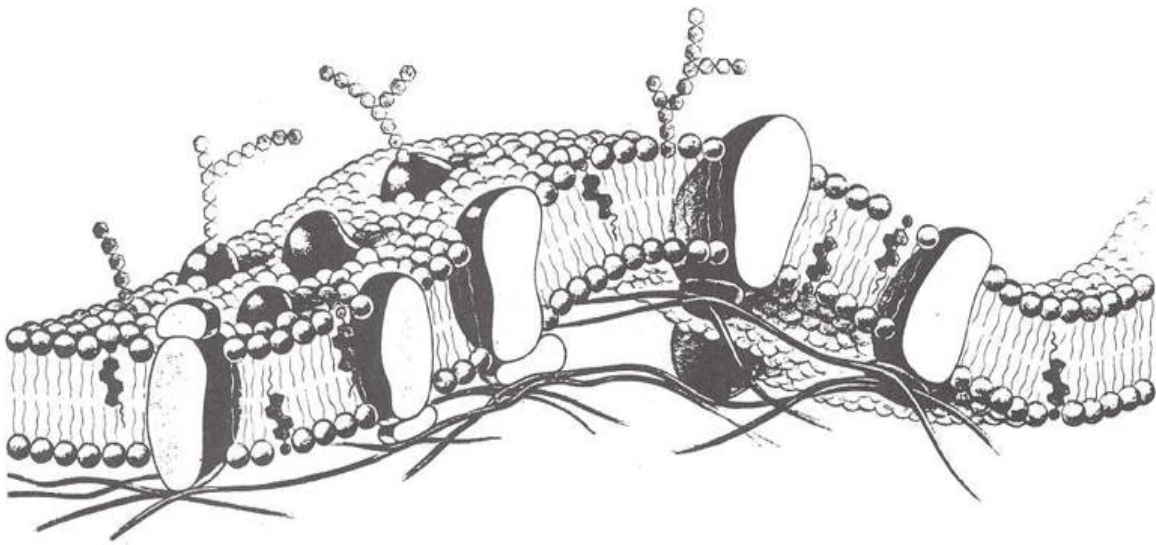


Fig. 1.4

- (d) Describe the properties of the phospholipid bilayer. [2]

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(e) Explain the significance of transport proteins in found in the membrane. [3]

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Question 2 [12 marks]

Fig. 2.1 below shows DNA replication in an organism.

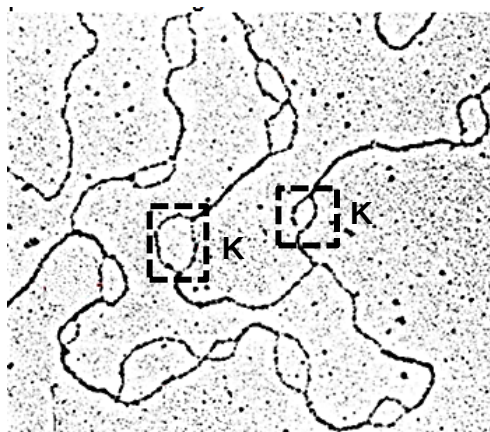


Fig. 2.1

(a) (i) With reference to **Fig. 2.1**, state, with a reason, if this organism is eukaryotic or prokaryotic. [1]

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(a) (ii) Explain the role of structures **K** in the cell. [1]

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- (b) (i) The DNA replication at each replication fork is sometimes described as ‘*asymmetrical*’ replication as there are differences in the way the daughter strands are being synthesized. State two such differences. [2]

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- (b) (ii) Suggest **two** reasons for these differences in (b) (i). [2]

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Telomeres undergo characteristic length changes over time in normal somatic and germline cells, and in premature ageing syndromes. In contrast to germ cells, which have high telomerase activity (indicated by the plus symbol on the graph), most somatic cells possess low or absent telomerase activity (indicated by the minus symbol on the graph). Telomere shortening is thought to contribute to the onset of degenerative diseases that occur during human ageing, eg. heart failure; immunosenescence (infections); digestive tract atrophies (ulcers); infertility; reduced viability of stem cells; reduced angiogenic potential; reduced wound-healing; and loss of body mass. In addition, several human premature ageing syndromes show an accelerated rate of telomere shortening, therefore resulting in an early onset of ageing-related pathologies. Examples are Ataxia telangiectasia (**ATM**); Werner syndrome (**WRN**); Bloom syndrome (**BLM**), etc.

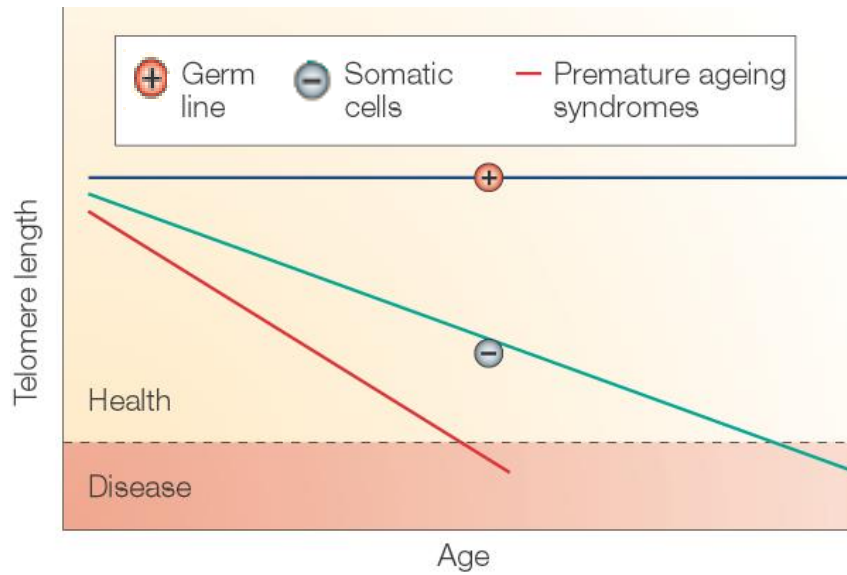


Fig. 2.2

(d) With reference to **Fig. 2.2**,

(i) explain the role of telomeres. [2]

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(ii) explain what happens when telomeres reach a critical length. [2]

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(iii) account for the differences in telomere lengths in germ cells and somatic cells. [2]

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Question 3 [12 marks]

Fragile X Syndrome (FXS), the most common inherited form of mental retardation and autism, which is caused by inactivation of the *Fmr1* gene, which encodes the Fragile X Mental Retardation Protein (FMRP). In a normal person, FMRP regulates the synthesis of synaptic proteins. **Fig 3.1** shows how patients with FXS have non-functional FMRP, resulting in accelerated synaptic protein synthesis that leads to abnormal synaptic function and intellectual disability.

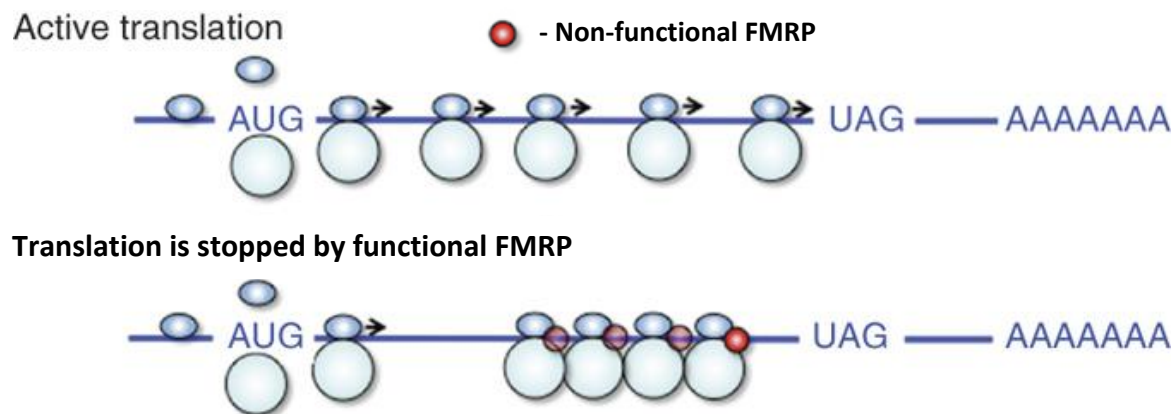


Fig. 3.1

With reference to **Fig. 3.1**,

(a) (i) state the level of control of gene expression and role exerted by FMRP. [1]

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(ii) state the role played by FMRP. [1]

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(iii) suggest how FMRP may regulate gene expression at the level stated in (a)(i) in a normal person. [2]

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(iv) suggest the advantage of this level of control over that of an earlier stage of gene expression. [1]

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Excessive synthesis of synaptic proteins, which causes FXS is due to inactivation of FMRP. The sequences of both normal and mutated FMRP are reflected in the figure below.

Functional FMRP	His – arg – leu – thr – pro – glu – ser –lys.....
Mutated FMRP	His – arg – leu – thr – pro – val – ser –lys.....

Fig. 3.2

(b) With reference to **Fig. 3.2**, explain how a mutation in the FMRP could result in increased protein synthesis that causes Fragile X syndrome. [3]

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- (c) In a particular mutation, one of the bases in the FMR1 gene sequence was replaced by another base. However, the FMR1 protein remained functional. Suggest two reasons how this may be possible. [2]

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- (d) Describe two ways in which accuracy in translation is achieved. [2]

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Question 4 [12 marks]

The following shows an electron micrograph of a bacteria cell infected by foreign particles.

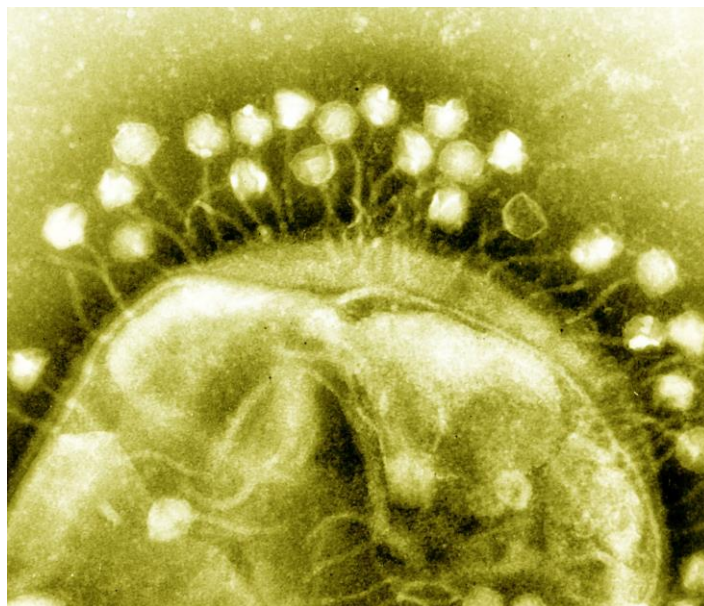


Fig. 4.1

The image was enlarged and the electron micrograph of a single particle is shown below.

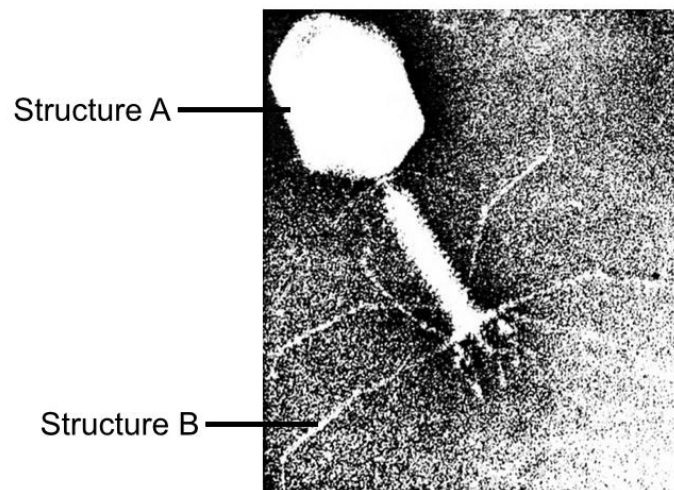


Fig. 4.2

(a) Identify this foreign particle. [1]

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(b) Identify the structural components labeled on **Fig. 4.2**. [1]

A:

B:

(c) Describe how the foreign particle enters the bacteria as shown in **Fig. 4.1**. [2]

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(d) Outline the function of **structure A**. [3]

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The following experiment was carried out to investigate the effect of bacteriophage and antibiotic on *E. coli*.

Three circular filter papers were soaked with either bacteriophage or antibiotics and placed on an agar plate, labeled Plate I, with a lawn of *E. coli*. The plate with the filter paper was then incubated for 24 hours and rings of clear zones were observed (**Fig. 4.3a**). These zones represent bacteria killed.

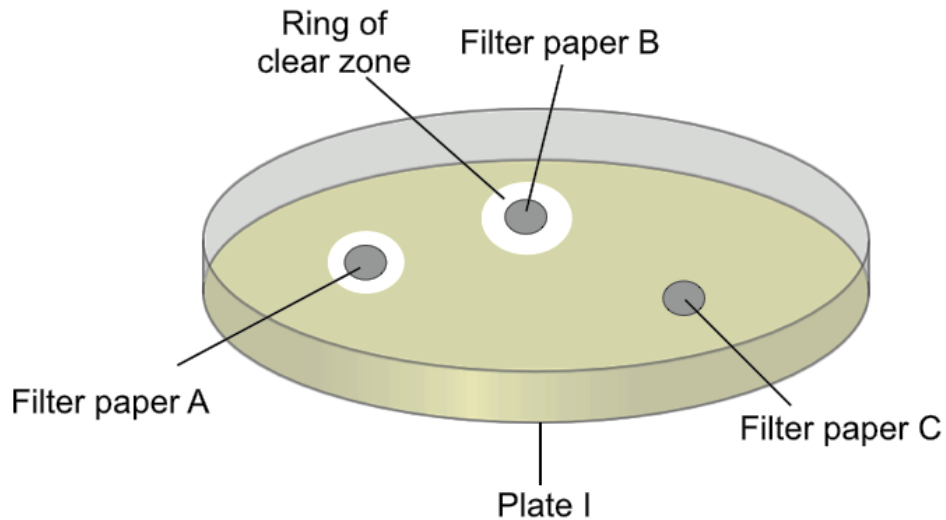


Fig. 4.3a

The same experiment was repeated on a second agar plate containing a different strain of *E. coli* (**Fig. 4.3b**).

Filter Paper A	Ampicillin
Filter Paper B	T4 bacteriophage
Filter Paper C	Lambda phage

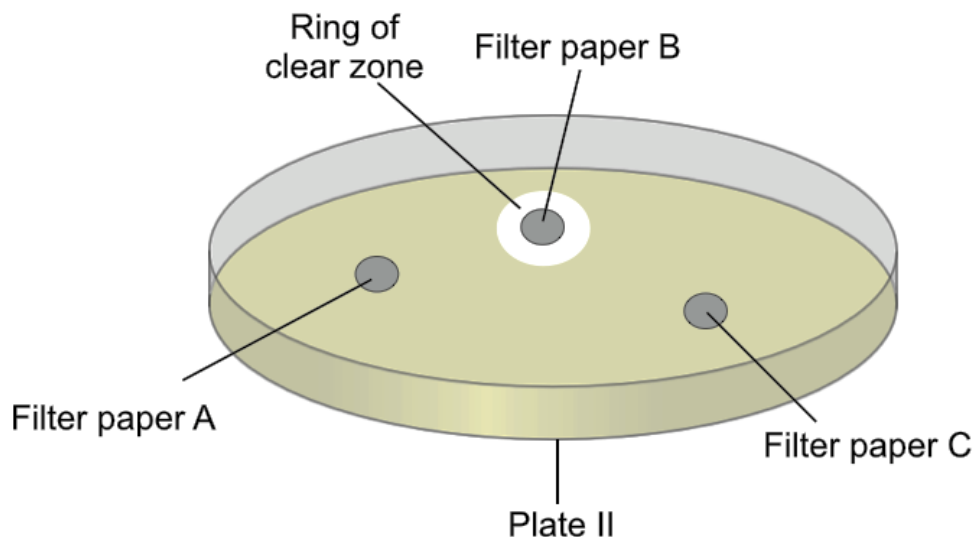


Fig. 4.3b

(e) With reference to **Fig. 4.3a and 4.3b**,

(i) identify the plate that will contain ampicillin resistant *E. coli*. Explain your answer. [1]

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(ii) account for the differences observed for filter papers B and C after 24 hours of incubation. [3]

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Phage therapy is the therapeutic use of bacteriophages to treat pathogenic bacterial infections. Phage therapy can be very effective in certain conditions and has some unique advantages over antibiotics.

(f) Suggest which of the two bacteriophages – T4 phage or lambda phage– will be more suitable for use in phage therapy. Explain your answer. [1]

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Question 5 [10 marks]

In cattle, coats may be solid white, solid brown, or beige. When true breeding solid whites are mated with true-breeding solid brown, the F1 generation consists of all solid white individuals. Matings among the F1 generation have resulted in the following ratio:

23 solid white
6 beige
3 solid brown

(a) Draw a genetic diagram of the above described cross. [4]

(b) Explain the genetic basis of this type of inheritance. [2]

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The chi-squared (χ^2) test was performed on the F₂ data giving a calculated value for χ^2 of 4.73.

Degrees of freedom	Probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

(c) State what conclusion may be drawn regarding the observed F₂ data. [2]

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In a particular variety of tomato plant, the allele for red fruit colour (**A**) is dominant over the allele for orange fruit (**a**) and the allele for green base when ripe (**B**) is dominant over the allele for no green base when ripe (**b**).

Two students, Eugene and Cornelius crossed plant with red fruit and green base when ripe with pure bred plant with orange fruit and no green bases when ripe. The phenotypes of 50 offspring of each of Eugene's and Cornelius's crosses were recorded and are shown in **Table 5.1**.

	Phenotypes of offspring of test crosses			
	Red fruit with green base	Red fruit with no green base	Orange fruit with green base	Orange fruit with no green base
Eugene's cross	23	4	3	20
Cornelius's cross	3	21	23	3

(d) Using the symbols given, state the genotypes of the tomato plants in Eugene's and Cornelius's crosses. [2]

Eugene's cross:

Cornelius' cross:

Question 6 [12 marks]

Ras is a GTP-binding protein that helps to broadcast signals from the cell surface to other parts of the cell to stimulate cell proliferation. If Ras function is inhibited, the cell proliferation responses normally induced by the activated receptors do not occur. Conversely, if a hyperactive mutant, Ras protein, is introduced into some cell lines, the effect on cell proliferation is the same as that induced by the binding of ligands to cell surface receptors. This hyperactive product of mutant ras gene promotes the development of cancer. **Fig. 6.1** below shows the sequence of events in a Ras cell cycle-stimulating pathway.

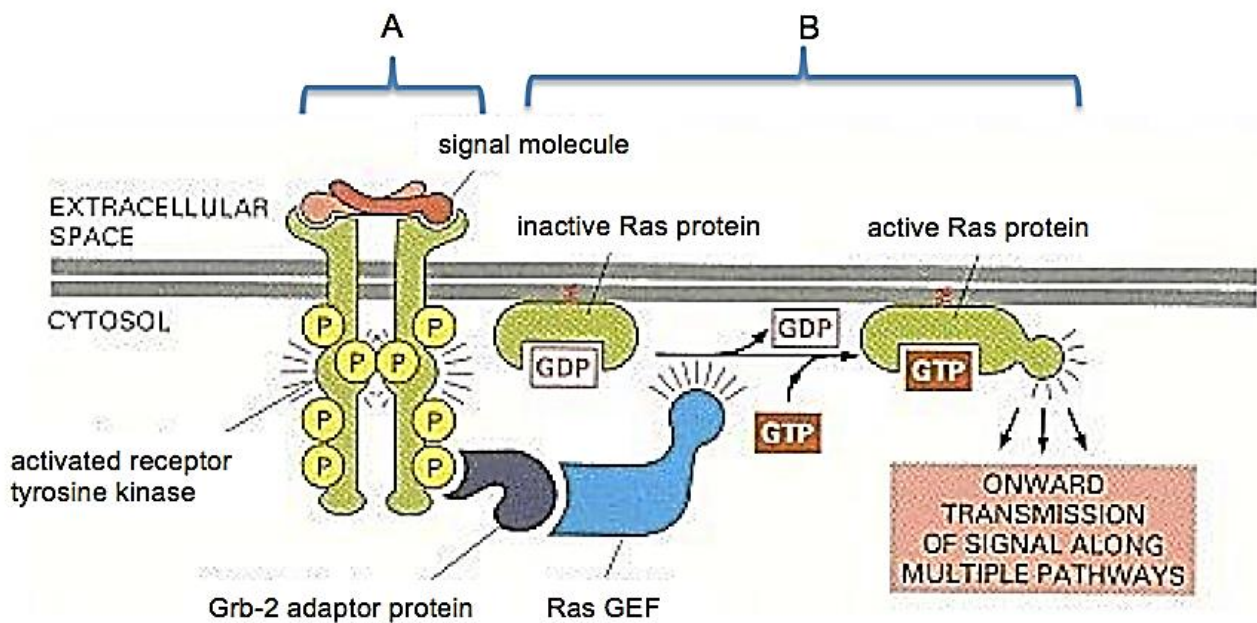


Fig. 6.1

(a) With reference to **Fig. 6.1**,

(i) briefly describe the **stages A and B**. [4]

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(ii) observe that Ras protein is a plasma membrane bound G-protein. Suggest how the activated G-protein can relay the signal from the ligand to activate cellular responses. [2]

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The receptor for epidermal growth factor (EGF) can be activated by a deletion that removes part of its extracellular domain. These mutant receptors are able to form active dimers even in the absence of EGF, and thus they produce a stimulatory signal inappropriately.

Besides mutation to the gene that codes for the extracellular domains of the receptors, 30% of human tumors have a hyperactive Ras mutation.

(b) Suggest how mutation in the Ras gene may bring about cancer. [3]

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(c) Briefly explain why the development of cancer is a multi-step process. [3]

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Question 7 [10 marks]

Soil salinity is one of the major environmental stresses causing a significant decrease in productivity in world culture agriculture as it can inhibit the growth of plants. Salinity refers to the occurrence of salts in ground water. Ions contributing to the salinity include Na^+ , K^+ , Mg^{2+} and Cl^- .

There are a few species of plants of genus *Arabidopsis* that are closely related to each other showing differences in ability to transport these ions out of the cell. They have all arisen from a common ancestor, but have evolved to live in different soil conditions.

The molecular basis for variation in the type of ions being transported out of the cell appears to be due to HKT gene. HKT gene codes for an ion channel known to transport these salts out of the cell (see **Fig. 7.1**). Some HKT only transport a specific ion while others can transport different ions simultaneously (see **Table 7.2**).

Based on the molecular information of all genes found in *Arabidopsis*, it has been studied that a common ancestor of *Arabidopsis* plants, *A.dentata*, contains HKT1 which codes for a protein that transports only Mg^{2+} . (see **Fig. 7.3**)

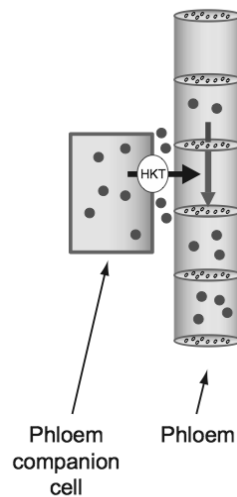


Fig. 7.1

Table 7.1

Salt tolerance of <i>Arabidopsis</i> plants	Type of soil salinity that it cannot survive in	Types of HKT present
High tolerance	Able to survive in all types of soil salinity	HKT 3 - transports only Na^+ , K^+ and Cl^-
Moderate tolerance	Soil rich in magnesium and potassium	HKT 2 – transports only Cl^- and Na^+
Sensitive	Soil rich in sodium, potassium and chloride	HKT 1 – transports only Mg^{2+}

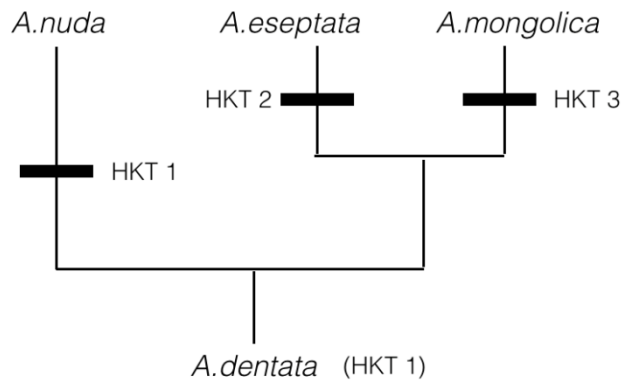


Fig. 7.2

(a) With reference to **Table 7.1** and **Fig. 7.2**,

- (i) state the environments that *A. nuda*, *A. eseptata* and *A. mongolica* are likely to have evolved in. [1]

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- (ii) explain how the molecular evidence described supports Darwin's theory of natural selection. [4]

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- (iii) identify two pairs of *Arabidopsis* species that each shows close relatedness. Explain your answer. [2]

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In a separate study on other Eukaryotes, neutral theory was put forth in accounting for much of the evolutionary changes that occur at the molecular level. The molecular clock for fibrinopeptides (peptides that aid in blood clotting), hemoglobin, cytochrome c and histone H4 was constructed to study the evolutionary rates of these proteins (see **Fig. 7.3**).

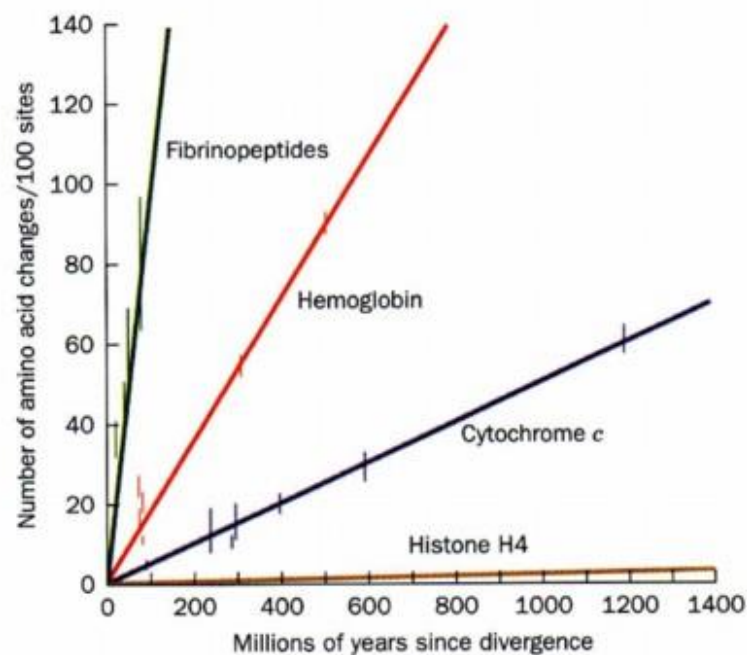


Fig. 7.3

- (b) Explain the basis of a molecular clock with respect cytochrome c. [1]

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(c) Account for the different evolutionary rates between fibrinopeptides and cytochrome c. [2]

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Section B [20 marks]

Answer **only ONE** question.

Write your answers on the separate answer paper provided. Your answers should be illustrated by large, clear labeled diagrams, where appropriate. Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

Question 8

- (a) Describe the structure of a lac operon and explain how it may be regulated. [9]
- (b) Describe the process and significance of binary fission in bacteria. [4]
- (c) Describe the process and significance of transformation and transduction in bacteria. [7]

Question 9

- (a) Compare and contrast between oxidative phosphorylation and photophosphorylation. [8]
- (b) Describe the structural adaptations of mitochondria for its function in ATP synthesis. [6]
- (c) Describe the role of NAD⁺ in aerobic respiration. [6]